

Rounak Sonavane

AI Systems Engineer | Intelligent Automation | Robotics | Computer Vision
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Profile

AI Systems Engineer passionate about designing and building end-to-end intelligent systems that combine AI, robotics, backend engineering, and intelligent automation. Experienced in architecting production-oriented applications involving LLMs, computer vision, autonomous workflows, and distributed system integration. Enjoy solving complex engineering problems through scalable software architecture and modern AI technologies.

Tech Stack

Languages	Python, C, C++, JavaScript, SQL
AI & Machine Learning	PyTorch, TensorFlow, LangChain, LangGraph, RAG, YOLO, OpenCV, MediaPipe, InsightFace
Backend	FastAPI, Flask, Express, REST APIs, Node.js
Databases	PostgreSQL, MongoDB, DuckDB, FAISS, ChromaDB
Robotics	ROS2, MoveIt
DevOps & Tools	Git, Docker, Linux
Automation	n8n

Projects

MindDesk — Multimodal Agentic AI Platform

Python, Flask, React, Ollama, PostgreSQL, MongoDB, FAISS, ChromaDB, YOLO, InsightFace github.com/Rounak7721/MindDesk

- Architected a modular plan-and-execute AI orchestration engine coordinating 17 specialized tools through a decoupled tool registry, enabling autonomous multi-step task execution
- Designed a document intelligence pipeline supporting semantic retrieval across 4 document formats (PDF, DOCX, XLSX, CSV) using hybrid vector search with FAISS and ChromaDB
- Engineered a multimodal AI platform integrating computer vision, speech processing, document intelligence, and IoT automation through a unified orchestration framework with model routing and monitoring

Agentic Lead Qualification Automation

n8n, LLMs, Airtable, Google Sheets, Slack

github.com/Rounak7721/ai-lead-qualification-system

- Built automated lead capture, qualification, scoring and routing system with stateful workflow orchestration
- Implemented LLM-based lead evaluation and follow-up automation with resilient workflow execution
- Designed hybrid cloud and local deployment support for privacy and cost efficiency

Virtual Mouse — Gesture-Controlled HCI System

Python, OpenCV, MediaPipe

github.com/Rounak7721/virtual-mouse-gesture-controlled-hci

- Built gesture-controlled human-computer interaction system using hand landmark tracking
- Implemented cursor movement, clicks, scrolling and drag interactions
- Designed for touchless interaction and accessibility use cases

IRIS — Intelligent Robotic Perception and Manipulation System (Ongoing)

Python, C++, ROS2, MoveIt, OpenCV, YOLO

github.com/Rounak7721/iris

- Designing and developing a custom 6-DOF robotic manipulation platform integrating ROS2, MoveIt, computer vision, motion planning, and autonomous task execution
- Integrating YOLO and dual-camera perception with MoveIt for real-time object tracking and manipulation

Experience

AI Intern — Ramkrishna Foundry Pvt. Ltd.

Jan 2025 – Apr 2025

- Developed a company website using Vite, JavaScript and Express with an integrated LLM-powered chatbot
- Automated contact form and data capture workflows using Google Sheets integration and web analytics tools
- Evaluated and prototyped LLM-powered features for business workflows and internal automation

Education

B.Tech in Artificial Intelligence & Machine Learning

Sanjay Ghodawat University

Additional

Languages: English, Hindi, Marathi

Open To: AI Systems Engineer, Intelligent Automation, Robotics Software, Applied AI, Computer Vision